

ENGINEERING CONCEPTS VISUALIZER

Project Report

A Web-Based Visual Learning Platform for Engineering Concepts

Project Areas

Signals | Circuits | Mechanics

Developed by

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Technologies: Java • JSP • MySQL

1. Introduction

Engineering Concepts Visualizer is a web-based educational application designed to help students understand difficult engineering concepts through visual and interactive learning resources. The platform focuses on Signals, Circuits, and Mechanics and provides a structured environment for students, faculty, and administrators.

Instead of depending only on traditional text-based explanations, the system organizes concepts with visual resources such as charts, animations, 3D visualizations, videos, and documents. This makes the learning process more engaging and easier to understand.

2. Problem Statement

Students often find engineering subjects difficult when concepts are presented mainly through theory, formulas, and static diagrams. Some concepts are easier to understand when students can see how values, shapes, graphs, and components behave.

The project addresses this problem by providing a single web platform where engineering concepts can be explored using visual learning materials and where student learning activity can be monitored.

3. Objectives

- Make difficult engineering concepts easier to understand through visual learning.
- Provide organized learning content for Signals, Circuits, and Mechanics.
- Allow faculty to add and manage educational concepts and resources.
- Allow administrators to review and manage concepts, users, and feedback.
- Allow students to track learning progress and provide feedback.
- Create a structured and role-based learning platform.

4. Proposed Solution

The proposed system provides separate modules for students, faculty, and administrators. Students can browse subjects and concepts, open learning resources, complete concepts, and submit feedback or ratings. Faculty members can add and update concepts and monitor student progress.

Administrators can manage users, subjects, concept approval, feedback, and reports.

5. System Modules

5.1 Student Module

- Student dashboard for accessing learning content.
- Browse concepts by subject.
- View concept descriptions and learning resources.
- Access charts, animations, 3D visualizations, videos, and documents.
- Mark concepts as completed and maintain progress.
- Submit feedback and ratings.

- View student profile and learning-related information.

5.2 Faculty Module

- Faculty dashboard for managing teaching content.
- Add new engineering concepts.
- Attach or reference learning resources.
- Edit and update concepts.
- View concepts created by the faculty member.
- Monitor student progress.
- View student feedback.

5.3 Admin Module

- Admin dashboard for overall system management.
- Manage students and faculty accounts.
- Manage subjects.
- Review, approve, reject, and update concept status.
- View feedback and reports.
- Manage user and concept information.

6. Key Features

- Role-based Student, Faculty, and Admin modules.
- Subject-based organization for Signals, Circuits, and Mechanics.
- Interactive charts and visual learning pages.
- Animation and 3D visualization pages.
- Video and document resources.
- Student progress and concept completion tracking.
- Concept ratings and feedback.
- Faculty concept management.
- Administrative concept approval and user management.
- MySQL-based data storage.

7. Technologies Used

Java: Used for application-side programming and server-side functionality.

JSP: Used to build the dynamic web pages and user interfaces for the application modules.

MySQL: Used to store users, subjects, concepts, feedback, ratings, and learning progress.

8. System Workflow

- User opens the web application and signs in or registers.
- The system identifies the user's role.
- The user is directed to the appropriate Student, Faculty, or Admin module.
- Students select subjects and access available concepts.
- Faculty members add or update concepts and learning resources.
- Administrators review and manage concepts, users, and feedback.
- Student completion, ratings, feedback, and progress are stored in the database.

9. Database Overview

The project includes a MySQL database named **eng**. The database dump included with the project contains the following tables:

- **concep** – stores concept information, subject/faculty references, resources, and status.
- **concept_feedback** – stores feedback associated with students and concepts.
- **concept_rating** – stores student ratings for concepts.
- **concepts** – stores concept-related data used by the application.
- **feedback** – stores student feedback records.
- **student_concept_status** – stores the status of a student's concept activity.
- **student_progress** – stores student learning progress.
- **subjects** – stores available subjects.
- **users4** – stores user account information used by the system.
- **users5** – stores additional user-related information present in the database.

10. Project Structure

The application follows a web-project structure with Java source files and JSP-based web pages. The main source is organized under **src/main**, while the database backup is stored separately in the **database** folder.

Folder / File	Purpose
src/main/java	Java application classes and server-side logic
src/main/webapp	JSP pages, web resources, and application modules
src/main/webapp/student	Student module pages
src/main/webapp/faculty	Faculty module pages
src/main/webapp/admin	Admin module pages
src/main/webapp/charts	Chart and visualization pages
src/main/webapp/animations	Subject animation pages
database	MySQL database backup
README.md	Project documentation for GitHub

Folder / File	Purpose
LICENSE	Project license information

11. Testing

The application can be tested module by module to verify that each role can access the correct functionality and that information is stored and retrieved correctly.

- Registration and login validation.
- Role-based navigation and access.
- Adding, editing, and viewing concepts.
- Admin approval and status updates.
- Student concept completion and progress updates.
- Feedback and rating submission.
- Database insertion and retrieval.
- Opening visual learning resources.

12. Advantages

- Makes engineering learning more visual and engaging.
- Combines multiple learning resources in one platform.
- Separates responsibilities using Student, Faculty, and Admin modules.
- Allows faculty to manage learning content.
- Provides progress and feedback information for learning improvement.
- Keeps project data organized in a centralized database.

13. Limitations

- The current system is primarily designed as a web application.
- The quality of learning depends on the resources provided for each concept.
- Some visualizations are implemented as dedicated pages rather than as a single reusable visualization engine.
- The current version does not include a dedicated mobile application.

14. Future Enhancements

- Personalized learning recommendations.
- Mobile application support.
- Multilingual learning support.
- Personalized learning paths.
- Advanced learning analytics.

- Gamification and achievement badges.
- Cloud deployment and scalable hosting.

15. Conclusion

Engineering Concepts Visualizer provides a practical approach to improving the understanding of engineering subjects through visual and interactive learning. By combining concept management, learning resources, progress tracking, feedback, and role-based access, the application creates a structured learning environment for students and faculty.

The project also demonstrates how a Java and JSP web application can be connected with a MySQL database to build a complete educational platform.

16. GitHub Repository Contents

The GitHub repository contains the application source code, database backup, project documentation, README, and license. Screenshots can be added later to the README and documentation to provide a visual demonstration of the application.

Project: Engineering Concepts Visualizer

Developer: Davala Madhuri

Primary Technologies: Java, JSP, MySQL